Distributionally Robust Learning for Social Program Participation Prediction: A Cross-Department Evaluation in Peru

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Abstract. Prediction models of social-transfer participation may have different errors in geographical areas not represented during development. We assessed reported participation in Pensión 65 and Juntos using the 2025 National Survey of Budget Programmes (ENAPRES 2025). The study included 11,354 households with an older adult and 17,671 with a child or adolescent. Six primary prediction specifications were evaluated by leaving out each of 25 Peruvian departments in turn; the asset principal component was estimated separately within every training partition. Group distributionally robust optimization (Group DRO) was selected against the largest source-validation departmental loss. CatBoost had the lowest pooled log-loss for Pensión 65 (0.3217). For Juntos, equal-department neural training and CatBoost had pooled losses of 0.2959 and 0.2962, respectively (paired difference -0.00024, 95% interval [-0.00211, 0.00169]). Relative to neural empirical risk minimization, Group DRO reduced maximum departmental loss from 0.6715 to 0.6161 for Pensión 65 and from 0.6466 to 0.6010 for Juntos, while increasing pooled loss and maximum entropy-normalized departmental loss in both tasks. Departmental calibration errors persisted despite small nationally pooled biases. The findings distinguish pooled accuracy, extreme departmental loss and local calibration. Further studies across survey years and independent national settings could determine the stability of these differences.

Keywords: Geographic generalization · Social programmes · Group DRO · Weighted principal components · ENAPRES 2025 · Calibration

1 Introduction

Predictive models for social-assistance outcomes are frequently evaluated by losses averaged across nationally pooled observations. Aggregate performance can conceal geographically concentrated errors when programme coverage, household

characteristics and outcome prevalence vary across jurisdictions. Withholding a complete geographical unit provides a direct assessment of predictive performance where no households from that unit contributed to model development [22,13,21].

Peru's Pensión 65 and Juntos programmes permit this question to be examined in two distinct populations. Pensión 65 provides non-contributory income support to older adults in extreme poverty. Juntos provides transfers to poor households conditional on commitments associated with health and education [18,17]. They differ in age composition, programme conditions and geographical patterns of reported receipt. The 2025 National Survey of Budget Programmes (ENAPRES 2025) records both outcomes alongside household demographic and material characteristics. Evaluating the programmes under a common territorial protocol allows differences in model performance to be studied across two substantively distinct transfer settings.

Earlier research has compared machine-learning and proxy-means-test predictions for social assistance [16], evaluated alternative data for transfer targeting [1], examined errors associated with biased beneficiary data [6], and studied geographic targeting or transfer-programme classification in Peru [4,15]. The present outcome is household-reported programme receipt, rather than poverty status, eligibility or the causal effect of a transfer. Its methodological focus is the behavior of fitted probabilities when an entire department is omitted from both training and hyperparameter selection. This differs from random household validation and allows a comparison of national loss, the maximum loss across departments, loss relative to local outcome entropy and calibration.

We compare a prevalence baseline, penalized logistic regression, gradient boosting and three neural training objectives, including group distributionally robust optimization (Group DRO) [23,8]. The asset component is re-estimated within each training partition. Group DRO is selected by the maximum validation-department loss, while a department-balanced logistic model provides a comparison of geographical weighting outside the neural architecture. We examine how these specifications rank under pooled, maximum departmental and local-entropy-normalized loss, as well as their calibration within departments.

2 Survey Data and Analytic Populations

2.1 Households and participation indicators

The analysis used the 2025 National Survey of Budget Programmes (Encuesta Nacional de Programas Presupuestales, ENAPRES 2025), conducted by Peru's National Institute of Statistics and Informatics (INEI) [11,10]. The study included 36,184 distinct households from 24 departments and the Constitutional Province of Callao, defining 25 geographical evaluation units. Household identifiers combined the survey year, month, department, conglomerate, dwelling and household codes; the month field distinguished records in the household key. The binary

outcome recorded whether at least one household member reported receiving each programme benefit.

The Pensión 65 sample comprised households with at least one member aged 65 years or older. The Juntos sample comprised households with at least one member aged 0–17 years. These restrictions defined two programme-specific prediction populations, with some households included in both. The outcomes were indicators of household-reported receipt. Sample sizes, observed positive counts and weighted prevalence are given in Table 1.

Table 1. Analytic populations in ENAPRES 2025. Weighted prevalence and Kish effective sample sizes use household expansion weights; counts are unweighted.

Programme	Households	Positive	Weighted (%)	Approx. clusters	Kish n_{eff}
Pensión 65	11,354	4,531	26.86	3,810	4,732
Juntos	17,671	3,341	14.90	4,262	7,365

Analyses used the positive household expansion weights, which summed to approximately 3.09 million in the Pensión 65 sample and 4.06 million in the Juntos sample. The weight-only Kish effective sample sizes are reported in Table 1 [12]. Variance estimation used approximate primary sampling units assembled from department and conglomerate identifiers.

The 25 Pensión 65 departmental samples contained 195–1,063 households, with weighted participation ranging from 5.4% to 75.7%. The corresponding Juntos samples contained 425–1,323 households and weighted participation ranged from 1.9% to 48.3%. The full departmental distribution appears in Supplementary Fig. S1.

2.2 Predictors, coding and exclusions

Both prediction tasks used the same 18 household predictors: 14 numeric or binary variables and four categorical variables (Table 2). The predictors comprise demographic composition, household structure, service access and an asset score derived from 17 ownership indicators. Categorical predictors were one-hot encoded for logistic regression and the neural models; CatBoost used native categorical encoding. The asset component, numerical standardization and categorical encoders were fitted separately within each training partition. Definitions and source fields appear in Supplementary Table S1.

Table 2. Operational definitions of the 18 household predictors used for both outcomes

Predictor	Operational definition	Scale
Head's age	Age of household head	Years
Household size	Number of household members	Count
Children aged 0–11	Number of members aged 0–11 years	Count
Adolescents aged 12–17	Number of members aged 12–17 years	Count
Adults aged 65+	Number of members aged 65 years or older	Count
Female members	Number of female household members	Count
Public piped water	Household connected to public piped-water supply (yes/no)	Binary
Public sewerage	Household connected to public sewerage network (yes/no)	Binary
Grid electricity	Household with grid electricity access (yes/no)	Binary
Internet access	Household with internet access (yes/no)	Binary
Digital device	Household with a digital device (yes/no)	Binary
Multigenerational household	Household classified as multigenerational (yes/no)	Binary
Disability present	At least one member with a disability	Binary
Asset index	Weighted first component of 17 asset indicators, estimated using training data	Score
Head's gender	Gender of household head (male or female)	Category
Head's education	Education category of the household head	4 levels
Head's mother tongue	Mother tongue of the household head	Category
Residence area	Urban or rural residence	2 levels

The asset index was calculated from the 17 binary asset fields using a survey-weighted principal-component analysis. Within each source training partition, indicators were centered and standardized, the first component was oriented toward higher asset ownership, and scores were standardized to source-weighted mean zero and variance one. The estimated centering, loadings and scaling were then applied to the validation or test department. Thus, the held-out department did not enter the estimation of the asset index. Variables with incomplete responses and identifiers of households or geographic evaluation units were excluded from the primary feature set; other programme indicators were also omitted.

3 Territorial Validation and Model Estimation

3.1 Prediction outcomes and risk measures

For programme $s \in \{\mathrm{P65}, \mathrm{J}\}$, let $\hat{p}_i^{(-d,m)}$ denote the predicted probability for household i from model m, fitted without observations from its department d.

The binary logarithmic loss is $\ell(y,p) = -y\log p - (1-y)\log(1-p)$. Weighted held-out departmental risk is

$$\widehat{R}_{d,m} = \frac{\sum_{i:D_i=d} w_i \ell(Y_i, \hat{p}_i^{(-d,m)})}{\sum_{i:D_i=d} w_i}. \tag{1}$$

Log-loss is a strictly proper scoring rule [7]. The household-weighted pooled score uses the expansion factors across all held-out observations. We also calculated the arithmetic mean of the 25 departmental losses, $\overline{R}_m = 25^{-1}\sum_d \widehat{R}_{d,m}$, and the largest departmental loss, $R_m^{\max} = \max_d \widehat{R}_{d,m}$. The three summaries assign different effective weights to geographical units.

For a prevalence-adjusted descriptive comparison, we divided each departmental loss by the entropy of its observed outcome prevalence. With $\hat{\pi}_d = \sum_{i:D_i=d} w_i Y_i / \sum_{i:D_i=d} w_i$ and $H_d = -\hat{\pi}_d \log \hat{\pi}_d - (1-\hat{\pi}_d)\log(1-\hat{\pi}_d)$, the largest ratio is

$$Q_m^{\max} = \max_{d=1,\ldots,25} \left\{ \frac{\widehat{R}_{d,m}}{H_d} \right\}. \tag{2}$$

The entropy H_d is the log-loss of a constant prediction equal to the observed departmental prevalence. The ratio therefore expresses model loss relative to a local prevalence benchmark. It is calculated from the held-out outcomes and used as a descriptive statistic alongside absolute log-loss.

3.2 Nested leave-one-department-out protocol

The outer evaluation comprised 25 folds. In each fold, all observations from one department formed the test set, and the remaining 24 departments formed the development set. Two inner `GroupKFold` partitions of development departments selected hyperparameters by household-weighted validation log-loss. All transformations, including the weighted asset PCA, were re-estimated within each inner training split and again on the 24 development departments before the outer prediction. The 25 test sets were combined after model fitting to obtain one out-of-department prediction per eligible household.

Department exclusion was used to estimate the loss associated with geographic transfer within the 2025 survey [22]. A secondary evaluation withheld approximate conglomerates while retaining all departments in the training pool. This supplementary exercise used fixed model settings, whereas the main evaluation included inner tuning. The resulting differences incorporate changes in both the partitioning and fitting protocols.

3.3 Competing algorithms and optimization objectives

Six specifications were evaluated. The baseline predicted the weighted source-department prevalence in each outer fold. Logistic regression used ℓ_2 regularization, with $C \in \{0.1, 1, 10\}$ selected by inner validation. CatBoost used 240

iterations, learning rate 0.05, leaf penalty six and candidate tree depths four and six [19]. The comparison was restricted to these candidate configurations.

The multilayer perceptron had hidden layers of 48 and 24 units, rectified-linear activation and a logistic output. Models were fitted for 95 full-batch AdamW epochs (learning rate 0.004); weight decay was selected from $\{0.002, 0.02\}$. Predicted probabilities were averaged across three seeds. Household-weighted empirical risk minimization (ERM) minimized $\sum_i w_i \ell_i / \sum_i w_i$; equal-department training minimized $D^{-1} \sum_{d \in \mathcal{D}_{\text{train}}} R_d(\theta)$, with $D = 24$. Group DRO updated source-department weights according to

$$\min_{\theta} \max_{q \in \Delta^{D-1}} \sum_{d \in \mathcal{D}_{\text{train}}} q_d R_d(\theta), \quad q_d^{(t+1)} \propto q_d^{(t)} \exp\{\eta R_d(\theta_t)\}. \tag{3}$$

Group DRO selected the weight decay $\lambda \in \{0.002, 0.02\}$ and update step $\eta \in \{0.04, 0.12\}$ jointly by minimizing the maximum weighted log-loss across the 24 source departments during inner geographical validation. In each inner split, evaluation departments remained absent from preprocessing and training. An additional sensitivity retained the previous selection by pooled validation loss; it is reported alongside a department-balanced logistic regression in the supplement. These specifications compare training and selection objectives within a fixed model architecture [23].

3.4 Discrimination, calibration and uncertainty

Secondary scores comprised the weighted Brier score, pooled and mean within-department ROC AUC, and weighted average precision [3,26,5,24]. Calibration-in-the-large was calculated as the factor-weighted mean predicted probability minus the factor-weighted observed proportion, both nationally and within departments. Pooled logistic recalibration provided calibration intercepts and slopes; decile-based reliability was summarized using equal-weighted-mass bins [27,9].

Paired pooled-loss differences were evaluated using 1,200 multinomial resamples of approximate conglomerates within departments. The eight comparisons were corrected for multiple testing using the Benjamini–Hochberg procedure [2]. A further 1,600 resamples assessed differences between maximum departmental losses, with the maximizing department reidentified in each replicate. All resampling estimates were conditional on the fitted out-of-department predictions [20,14].

4 Results

4.1 National and department-averaged performance

The models were evaluated on one held-out prediction per household, with 25 geographical folds for each outcome. CatBoost produced the lowest pooled weighted log-loss for Pensión 65 (0.3217), compared with 0.3551 for logistic

regression and 0.3412 for neural ERM. For Juntos, equal-department neural training, CatBoost and neural ERM had pooled losses of 0.2959, 0.2962 and 0.2965, respectively (Table 3); the differences between the first two were small relative to the sampling variation of the paired contrast. The pooled ROC AUC values for CatBoost were 0.914 and 0.867. Both outcomes benefited from household covariates relative to a source-prevalence prediction, although that baseline also varied across outer training folds.

Table 3. Out-of-department predictions pooled with household survey weights. AP denotes weighted average precision; $\overline{R}$ is the unweighted mean of 25 departmental log-loss values.

Task	Model	Log-loss	Brier	ROC AUC	AP	$\overline{R}$
Pensión 65	Source prevalence	0.6228	0.2139	0.212	0.175	0.6822
	Logistic L2	0.3551	0.1123	0.895	0.744	0.4382
	CatBoost	0.3217	0.1006	0.914	0.808	0.3990
	MLP–ERM	0.3412	0.1082	0.903	0.780	0.4211
	MLP–equal departments	0.3414	0.1083	0.904	0.781	0.4149
	MLP–Group DRO	0.3520	0.1124	0.900	0.771	0.4196
Juntos	Source prevalence	0.4371	0.1313	0.236	0.096	0.4773
	Logistic L2	0.3000	0.0936	0.864	0.502	0.3436
	CatBoost	0.2962	0.0922	0.867	0.517	0.3402
	MLP–ERM	0.2965	0.0927	0.867	0.509	0.3414
	MLP–equal departments	0.2959	0.0927	0.868	0.511	0.3411
	MLP–Group DRO	0.3050	0.0962	0.869	0.512	0.3528

The equal-department means for CatBoost were 0.3990 and 0.3402, exceeding its pooled household-weighted losses. This difference reflects the larger contribution of smaller departments to the macro-average. The source-prevalence model predicts a constant within each held-out department but uses a different constant across folds. Its within-department ROC AUC is therefore 0.5; pooled ROC AUC instead ranks these fold-dependent constants. Mean within-department discrimination and all 25 departmental losses are reported in the supplementary tables.

4.2 Paired comparisons of national log-loss

The CatBoost–logistic differences in pooled loss were -0.0334 and -0.0038 for Pensión 65 and Juntos. CatBoost–ERM differences were -0.0195 and -0.0003. Cluster-resampled percentile intervals and multiplicity-adjusted comparisons are reported in Table 4. In a separate exploratory comparison, equal-department neural training minus CatBoost for Juntos was -0.00024 (95% interval [-0.00211, 0.00169]; $p = 0.80$). This contrast does not establish a difference in pooled risk between the two methods. The paired resampling preserved the fitted probabilities and the geographical fold assignments; inference is conditional on these estimated prediction functions.

Table 4. Paired differences in pooled weighted log-loss. Intervals are percentile ranges from 1,200 resamples of approximate survey conglomerates within departments; BH correction is applied to eight comparisons. Negative values favour the first method.

Task	First − second	Difference	95% range	BH q
Pensión 65	CatBoost − logistic	−0.0334	[−0.0389, −0.0280]	< 0.001
Pensión 65	CatBoost − neural ERM	−0.0195	[−0.0234, −0.0155]	< 0.001
Pensión 65	Group DRO − neural ERM	+0.0108	[+0.0060, +0.0157]	< 0.001
Pensión 65	Equal-domain − ERM	+0.0003	[−0.0027, +0.0030]	0.864
Juntos	CatBoost − logistic	−0.0038	[−0.0056, −0.0018]	< 0.001
Juntos	CatBoost − neural ERM	−0.0003	[−0.0021, +0.0014]	0.850
Juntos	Group DRO − neural ERM	+0.0085	[+0.0042, +0.0128]	< 0.001
Juntos	Equal-domain − ERM	−0.0005	[−0.0018, +0.0006]	0.524

4.3 Maximum departmental loss and relative risk

Neural ERM attained maximum departmental losses of 0.6715 for Pensión 65 and 0.6466 for Juntos. With worst-department-selected Group DRO, the maxima were 0.6161 and 0.6010, corresponding to differences of -0.0554 and -0.0455. Table 5 also reports maximum log-loss divided by the binary entropy of the observed departmental prevalence.

Table 5. Worst held-out departmental log-loss ($R^{\max}$) and maximum local-entropy ratio ($Q^{\max}$) across the 25 evaluation units.

Task	Model	$R^{\max}$	Department	$Q^{\max}$	Department
Pensión 65	CatBoost	0.655	San Martin	1.014	Tumbes
	MLP–ERM	0.672	Tumbes	1.047	Tumbes
	MLP–equal departments	0.637	Tumbes	1.004	Callao
	MLP–Group DRO	0.616	Tumbes	1.113	Callao
Juntos	CatBoost	0.638	Huancavelica	1.288	Moquegua
	MLP–ERM	0.647	Huancavelica	1.289	Ucayali
	MLP–equal departments	0.638	Amazonas	1.328	Ucayali
	MLP–Group DRO	0.601	Huancavelica	1.695	Ucayali

Group DRO had lower loss than ERM in 10 of 25 Pensión 65 departments and 8 of 25 Juntos departments. The mean loss in the five highest-risk departments decreased from 0.6073 to 0.5685 and from 0.5738 to 0.5441, respectively. Conditional cluster-resampling intervals for the maximum-loss differences were [-0.1569, 0.0165] for Pensión 65 and [-0.0691, -0.0280] for Juntos. Thus, the Pensión 65 maximum-loss contrast was less precise than its observed direction alone would suggest.

The maximum departmental loss-to-entropy ratio increased from 1.047 to 1.113 for Pensión 65 and from 1.289 to 1.695 for Juntos. The worse relative-risk results occurred despite improvements in maximum absolute loss (Fig. 1). The entropy ratios use held-out outcome prevalence only in evaluation.

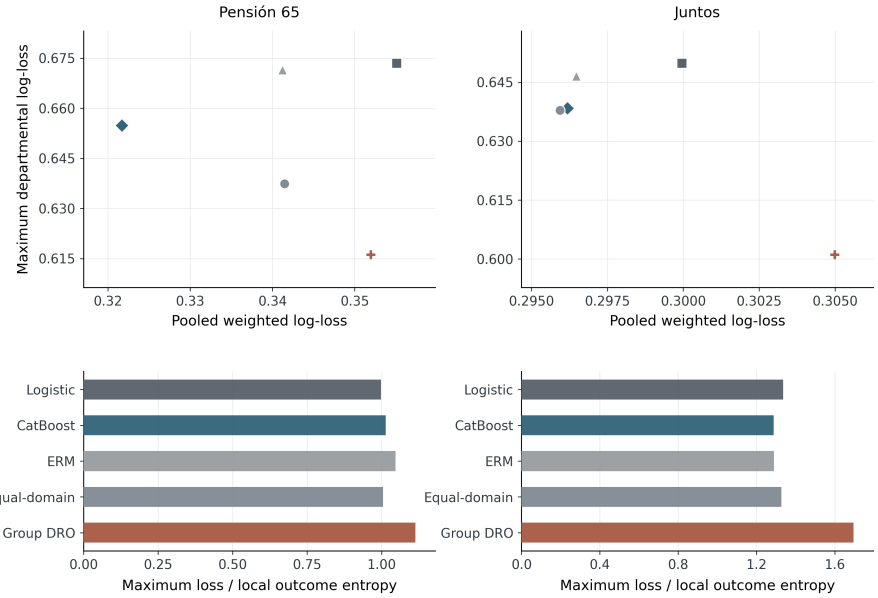


Fig. 1. Pooled and maximum departmental log-loss (upper panels) and the maximum ratio of departmental loss to observed local outcome entropy (lower panels). All scores use predictions from geographically held-out households.

4.4 Calibration across departments

CatBoost's pooled mean calibration biases were 0.04 and -0.21 percentage points, and weighted logistic recalibration slopes were 1.036 and 1.000. Weighted ten-bin expected calibration errors were 0.0102 and 0.0068. In contrast, its largest departmental underpredictions were -17.0 points in San Martin and -13.1 points in Huancavelica, respectively. The corresponding grouped-resampling intervals were [-23.4, -10.4] and [-18.8, -7.2] percentage points. Across the remaining departments, signs and magnitudes of calibration bias varied (Fig. 2); the full set of departmental estimates is in the supplement.

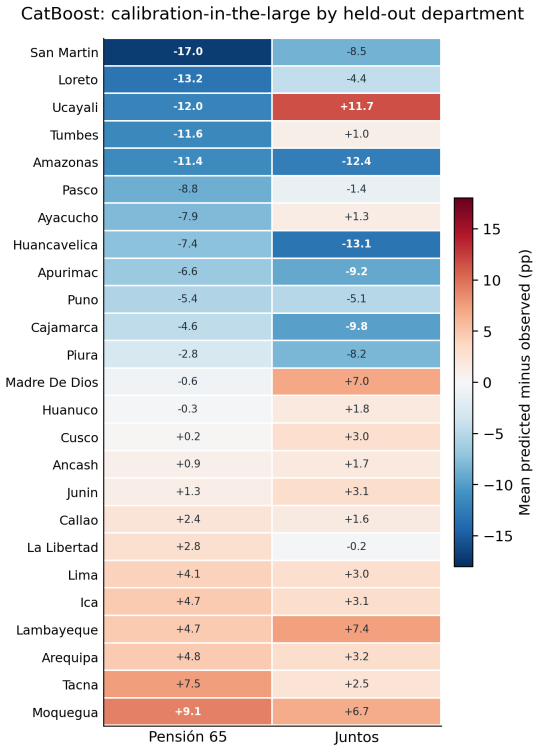


Fig. 2. CatBoost departmental calibration-in-the-large: weighted mean predicted probability minus weighted observed participation, in percentage points. The two outcome-specific populations differ and partially overlap.

4.5 Residence, natural region, and robustness checks

Table 6 compares weighted participation, log-loss and mean calibration bias across residence and natural-region subpopulations. Subgroup scores are partitions of the same national out-of-department predictions; regional models were not refitted for this table. Differences in participation prevalence contribute to differences in absolute log-loss. The local-entropy statistic addresses a distinct comparison with an outcome-specific constant benchmark.

Table 6. CatBoost subgroup outcomes. Participation is the weighted proportion reporting receipt, and bias is mean predicted minus observed participation (percentage points). Residence and natural region form separate partitions.

	Pensión 65			Juntos		
Subset	Prev. (%)	Loss	Bias (pp)	Prev. (%)	Loss	Bias (pp)
Urban	13.5	0.2821	+0.41	6.5	0.1996	−0.04
Rural	70.8	0.4518	−1.19	44.7	0.6380	−0.83
Coast	9.1	0.2355	+3.48	4.9	0.1671	+1.72
Highlands	50.2	0.3949	−2.41	26.8	0.4238	−2.22
Amazon	45.1	0.5490	−10.71	22.7	0.4505	−2.34

Under five-fold within-department conglomerate validation with fixed settings, CatBoost log-loss was 0.3161 for Pensión 65 and 0.2909 for Juntos. Compared with department exclusion (0.3217 and 0.2962), differences reflect both the partition scheme and the distinct fitting protocols. A fixed-specification sensitivity excluding the asset index produced pooled loss of 0.3418 and 0.3071, compared with 0.3217 and 0.2961 when retaining the fold-specific component. Additional analyses of training weights, natural-region inclusion, outcome sample restrictions, alternative geographical holdouts, covariate shift and seed variability are reported in the supplement.

4.6 Additional geographical reweighting comparisons

Group DRO selected by the maximum source-validation departmental loss had higher pooled loss than the earlier pooled-selected specification (0.3520 versus 0.3479 for Pensión 65; 0.3050 versus 0.2978 for Juntos), but lower maximum departmental loss (0.6161 versus 0.6195; 0.6010 versus 0.6151). As a non-neural reweighting comparison, equal total training weight per source department reduced the maximum loss of logistic regression from 0.6734 to 0.6189 for Pensión 65 and from 0.6498 to 0.6384 for Juntos. Complete paired contrasts and selected hyperparameters are given in the supplementary results.

5 Discussion

The evaluation produced different rankings under national and geographic loss summaries. CatBoost had the smallest pooled log-loss for Pensión 65 (0.3217), but its unweighted mean departmental loss was 0.3990. For Juntos, the corresponding CatBoost values were 0.2962 and 0.3402. The latter summaries give the 25 departments equal weight rather than weighting each household by its survey expansion factor. For Juntos, the observed difference between equal-department neural training (0.2959) and CatBoost (0.2962) was -0.00024, with a paired 95% interval spanning zero. The paired contrast does not establish a difference in pooled risk between these two models for Juntos. CatBoost had a pooled ROC

AUC of 0.867, compared with 0.868 under equal-department neural training. Separate calibration estimates are required to assess differences not captured by the discrimination and loss summaries.

Worst-department tuning addressed the discrepancy between the original Group DRO objective and its pooled-loss hyperparameter selection. Group DRO selected by maximum validation departmental loss achieved a maximum held-out loss of 0.6161 for Pensión 65 and 0.6010 for Juntos. The earlier pooled-validation selection gave 0.6195 and 0.6151, respectively, while yielding better national losses (0.3479 and 0.2978 versus 0.3520 and 0.3050). The contrast indicates that selection against source-department tail risk can further reduce observed test-domain extremes, but it need not improve aggregate prediction. In resampling comparisons against neural ERM, the interval for the Pensión 65 maximum-loss difference included zero; that for Juntos did not. In the held-out evaluations, Group DRO improved on ERM in 10 of 25 Pensión 65 departments and eight of 25 Juntos departments; reductions in the highest observed loss therefore did not imply widespread reductions in departmental error. The model-selection comparison is specific to the architecture and hyperparameter grid examined [23,8].

Equal weighting of training departments provides a distinct comparison with adversarial reweighting. The neural equal-department objective attained maximum losses of 0.6374 and 0.6379 for the two tasks, while ordinary neural ERM attained 0.6715 and 0.6466. To examine reweighting apart from neural optimization, a logistic regression using equal total weight per source department was also fitted with the regularization selected for conventional logistic regression. Its worst loss decreased from 0.6734 to 0.6189 for Pensión 65 and from 0.6498 to 0.6384 for Juntos. The first change coincided with a rise in pooled loss from 0.3551 to 0.3581; the second coincided with a small fall from 0.3000 to 0.2992. The Pensión 65 linear comparison thus produced a 0.0545 decrease in worst-department loss and a 0.0030 increase in pooled loss. The smaller Juntos changes indicate that the consequences of balancing department weights differed between the two analytic populations.

The normalized-risk and calibration analyses revealed additional differences. Although worst-department-selected Group DRO reduced absolute maximum loss, the maximum loss relative to local outcome entropy rose from 1.047 to 1.113 for Pensión 65 and from 1.289 to 1.695 for Juntos relative to neural ERM. The department with maximum Group DRO loss for Juntos was Huancavelica, whereas Ucayali had the highest loss relative to local entropy. For Pensión 65 the corresponding departments were Tumbes and Callao. Thus, normalizing by the observed participation rate changes which department is identified as most difficult. CatBoost's nationally pooled calibration bias was near zero for both programmes, while its largest departmental underpredictions reached 17.0 and 13.1 percentage points. Pooled calibration can therefore obscure opposing geographical biases [27,25].

The analysis concerns the conditional probability of reported programme receipt among households in the two specified age groups. At an aggregate level, these estimates allow comparisons of reported participation across territories. Departments with large differences between observed and predicted receipt could be prioritised for additional survey verification or local recalibration before interpreting coverage statistics. They do not estimate household poverty, administrative eligibility, unmet need or effects of programme receipt. Those outcomes require different labels, sampling frames and performance criteria. Pensión 65 and Juntos cover contrasting stages of the life course and different transfer arrangements; observing similar challenges in both strengthens the within-country comparison but is not an empirical test of cross-country transportability. The holdout protocol and risk measures are applicable to other multi-region surveys, yet their numerical performance must be established anew in those settings [13,1].

Several factors limit precision and scope. Both tasks draw on a single survey year, and the age-restricted populations partly overlap. The approximate-conglomerate bootstrap conditions on fitted predictions and does not account for the complete complex-sampling design or retraining variability. Selection by the largest validation-group loss can be sensitive to smaller source departments, and its behaviour under other architectures or optimization grids was not assessed. No separate national survey or administrative dataset was available for external validation. Consequently, the reported differences concern geographical transfer within the 2025 Peruvian survey, rather than future cohorts or social-protection systems in other countries.

6 Conclusion

In out-of-department prediction of reported participation in Pensión 65 and Juntos, pooled risk and the largest departmental loss did not identify the same preferred model. CatBoost attained the lowest pooled log-loss for Pensión 65 (0.3217). For Juntos, the difference between equal-department neural training (0.2959) and CatBoost (0.2962) was small and its paired 95% interval included zero. Selecting Group DRO against the worst source-validation department decreased the maximum loss from 0.6715 to 0.6161 for Pensión 65 and from 0.6466 to 0.6010 for Juntos relative to neural ERM. However, pooled loss rose from 0.3412 to 0.3520 and from 0.2965 to 0.3050, respectively, and the worst entropy-normalized ratio increased in both tasks. The department-balanced logistic comparison confirmed that the weighting of source territories can change upper-tail error without adversarial neural training.

Despite small pooled calibration biases, the largest CatBoost underpredictions were 17.0 percentage points in San Martin and 13.1 in Huancavelica. Pooled loss, the upper tail of departmental loss and local calibration therefore provide distinct assessments of geographic prediction. Further survey waves and independent geographical samples could establish how stable these differences are.

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